5. A lift is on the ground floor. If it goes 5 floors down and then moves up to 10 floors from there. Then in which floor will the lift be?

6. When Kala woke up, her body temperature was 102°F. She took medicine for fever. After 2 hours it was 2°F lower. What was her temperature then?

7. What number should be added to (–17) to get (–19)?

8. A student was asked to subtract (–12) from (–47). He got -30. Is he correct? Justify.

Objective type questions

9. $(-5)-(-18)=$ ____

(i) 23 (ii) –13 (iii) 13 (iv) –23

10. $(-100)-0+100=$ ____

(i) 200 (ii) 0 (iii) 100 (iv) –200

1.4. Multiplication of Integers

Situation 1

Ramani and Ravi are playing with pebbles in a heap. Ramani is adding some pebbles to the heap while Ravi removes some pebbles. First, Ramani adds 3 pebbles. Then she adds 3 more pebbles. She repeats this 2 more times. Can you say how many pebbles she added in all? Since adding pebbles is positive we can write $(+3)+(+3)+(+3)+(+3)=+12$ or $4\times(+3)=12$

The total number of pebbles added is 12.

Ravi takes out 5 pebbles three times. If we represent "taking out" as a negative integer then $(-5)+(-5)+(-5)=-15$ or we can simply put it as $(-5)\times 3=-15$. Ravi has removed 15 pebbles from the heap. This illustrates that multiplication of negative integers is also repeated addition just like positive integers or whole numbers.

We can draw a number line and visualise the above multiplication of integers as repeated addition.

$4\times 3=12$ (3 is added four times).

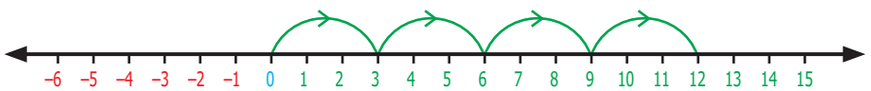


Fig. 1.18

$(-5)\times 3=-15$ ((–5) is added three times).

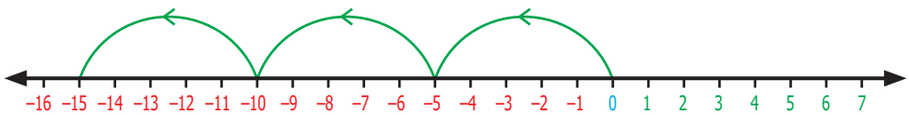


Fig. 1.19

We have no difficulty to understand that a positive integer like (+7) multiplied by another positive integer like (+8) is positive (+56). Also, a positive integer like +7 multiplied by a negative integer like –5 is –35 and $+5\times -7=-35$. But what about a negative integer

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